Multimedia Signal Processing - 2nd module

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February, 22 2011

1 Question - 10 pts

Consider a coding scheme that needs to transmit the position of a mouse pointer and mouse click events. You are given the following specifications:

- The screen resolution is 640×480 pixels.
- The coordinates (x, y) are expressed as integer values.
- There are three types of events e: left click (L), right click (R), idle (I).
- The mouse data $(x, y, e)_t$, where t denotes a time index, is updated at a frequency of 60Hz. An algorithmic delay of 1/30 seconds is tolerable.
- The following (rather unrealistic, though) statistics are given

$$\begin{aligned} p(L) &= 0.10 \\ p(R) &= 0.05 \end{aligned} \tag{1}$$

- Each triple $(x, y, e)_{t_i}$ is assumed (for simplicity) statistically independent from $(x, y, e)_{t_j}$, $t_i \neq t_j$.

Add additional assumptions, whenever needed, and do the following.

1. Design a lossless coding scheme (e.g. based on Huffman coding), and compute the overall average bitrate.
2. Discuss how to modify your design in order to take into account statistical dependencies between $(x, y, e)_{t_i}$ and $(x, y, e)_{t_j}$, $t_i \neq t_j$.

2 Question - 5 pts

Briefly illustrate transform coding. In particular, you need to discuss:

1. The application scenarios where it might be adopted;
2. How optimal bit allocation is performed;
3. The relationship with vector quantization.

3 Questions - 5 pts (each answer can be either TRUE or FALSE)

1. Consider an arithmetic coder and a discrete source X
   - [T] [F] The average codelength L is such that $L \leq H(X) - 1$
   - [T] [F] Source memory can be exploited by adaptively tracking the source statistics.
   - [T] [F] The arithmetic coder can be used only if X is a binary source
2. Consider a speech coding system
   - [T] [F] The short term predictor is useful when coding unvoiced speech
   - [T] [F] Quantization of reflection coefficients leads to instability at the decoder
   - [T] [F] The number of zeros of the whitening filter $A(z)$ depends on the number of formants below 4 kHz
3. Consider a block-based image coding system
   - [T] [F] Zonal quantization adaptively allocates the bitrate across the image
   - [T] [F] Decreasing the block size, the coding efficiency decreases
   - [T] [F] DC coefficients in different blocks are statistically independent
   - [T] [F] The PSNR between the original and the compressed image increases, as the bitrate increases

4 Question (Matlab) - 12 pts

A wireless multimedia sensor needs to send the greyscale image `marshall.pgm` to the base station. A radio channel is used at a data rate of 256 kbps.

- Find the transmission time of the uncompressed image.
- Find the minimum transmission time of the entropy-coded image (you can model pixel values as symbols of a discrete memoryless source)
- Given that the image needs to be delivered in less than 1 second, implement a PCM coding scheme that satisfy this constraint. Compute the PSNR of the compressed image with respect to the original.
- Implement a simple JPEG baseline (8×8 blocks, quality factor $q = 1$) using the DCT quantization matrix `Qjpeg.mat`. Compute the transmission time and the PSNR in this case. (Hint: you can estimate the rate using the ρ-domain model: $R = \theta(1 - \rho)$ where ρ is the fraction of quantized DCT coefficients equal to zero and $\theta = 5$).